

PAUL JOHN KITUI

✉ paulkitui@gmail.com ☎ 0703440488 📍 Nairobi, Kenya 🔗 My Portfolio, GitHub

🌐 [linkedin.com/in/paul-kitui-b058b1418](https://www.linkedin.com/in/paul-kitui-b058b1418)

📄 Professional Summary

Data & AI Solutions Engineer with experience across cloud data engineering, applied AI, technical delivery, product development, and analytics. Experienced in designing and building end-to-end data and AI solutions, including scalable data platforms, real-time pipelines, intelligent applications, backend services, and human-led AI orchestration workflows. Strong hands-on experience with Python, SQL, PySpark, Databricks, Google Cloud, BigQuery, FastAPI, LLMs, AI agents, and modern cloud-native architectures. Combines solution architecture, engineering, and business understanding to deliver secure, reliable, and production-ready systems that turn data and AI into practical business outcomes.

🔗 Career Highlights

- 7+ years across technology delivery, progressing from product development and project management into data platform engineering, analytics, and AI solutions.
- Architected a governed Databricks Lakehouse spanning Bronze, Silver, Gold, and Semantic layers, with automated data quality, governance, and BigQuery publishing.
- Engineered batch and real-time data pipelines on GCP using PySpark, Pub/Sub, Dataflow, BigQuery, and Cloud Run, including resilient event processing and observability.
- Built analytics and decision-support solutions, transforming curated data into semantic models, KPIs, dashboards, and actionable business insights.
- Led cross-functional technology initiatives from requirements and solution design through implementation, deployment, monitoring, and operational support.
- Delivered product and process improvements through data-driven decision-making, workflow optimization, and customer-focused solution development, including a product enhancement contributing approximately 15% revenue growth.

🧠 Technical & Professional Expertise

AI Engineering & Agentic Systems — LLM Applications • AI Agents • Human-in-the-Loop AI • Agentic Workflows • Tool / Function Calling • MCP • Prompt Engineering • Structured Outputs • Guardrails • State & Session Management • Agent Evaluation • Responsible AI

Cloud Data Engineering — Python • SQL • PySpark • Apache Spark • Databricks • Delta Lake • ETL/ELT • Batch Processing • Stream Processing • Data Pipeline Development • Data Quality Engineering

Solution & Data Architecture — Solution Architecture • Medallion Architecture • Event-Driven Architecture • Semantic Modelling • Data Platform Design • API Architecture • Integration Design • Scalability • Reliability • Security • Governance

Google Cloud & Data Platforms — BigQuery • Cloud Run • Pub/Sub • Dataflow • Cloud Storage • Firestore • Secret Manager • Cloud Build • Artifact Registry • Cloud Logging • Cloud Monitoring • IAM

Technical Delivery & Collaboration — Technical Project Management • Requirements Gathering • Stakeholder Management • Product Development • Agile/Scrum • Cross-Functional Leadership • Risk Management • Executive Reporting • Solution Delivery

📁 Employment

Sep 2026 – Present
Kenya

AI Engineer
WORKOPTi

- Design and architect production-grade human-led AI systems, defining orchestration patterns, agent workflows, control points, state/session handling, safety boundaries, and integration architecture for reliable AI execution.
- Build LLM-powered and agentic applications that transform unstructured information into structured Action Points™, using tool/function calling, prompt architectures, retries, stop conditions, guardrails, and mandatory human-in-the-loop review where required.
- Develop secure AI platform integrations and backend services using Python, FastAPI, MCP/equivalent tools, structured schemas, authentication boundaries, idempotency, and clear controls over what data can be exposed to models.
- Implement evaluation, observability, and operational readiness for AI systems, including tracing, regression testing, quality signals, auditability, human override, incident safeguards, and production controls needed to deploy and operate AI services safely at scale. AI Engineer JD Updated (1) AI Engineer JD Updated (1)

Sep 2024 – Aug 2026
Nairobi

Data Platform Engineer
Rulit Co. Limited

- Design and deliver end-to-end data and AI solutions, translating business requirements into scalable architectures, cloud platforms, intelligent applications, and analytics solutions.
- Build batch and real-time data platforms using Python, PySpark, Databricks, Google Cloud, BigQuery, Pub/Sub, and Dataflow, covering ingestion, transformation, modelling, validation, and analytics delivery.

- Develop AI and machine-learning solutions for decision support, predictive analytics, data quality, customer intelligence, and intelligent workflow automation, integrating models and LLM capabilities into practical business applications.
- Design solution architectures and backend services using FastAPI, APIs, event-driven patterns, and cloud services, ensuring systems are scalable, secure, maintainable, and suitable for production environments.
- Implement data quality, governance, reliability, and observability controls, including validation, lineage, audit logging, monitoring, exception handling, idempotency, and governed access to data.
- Manage solutions across the full engineering lifecycle, from requirements gathering and architecture through development, testing, deployment, monitoring, documentation, and continuous improvement.

Aug 2021 – Aug 2024
Nairobi

Technical Project Delivery Lead
CloudFactory

- Led end-to-end delivery of AI and data operations programs across engineering, operations, quality assurance, and client teams, ensuring successful execution against delivery milestones and quality standards.
- Engineered KPI dashboards and operational reporting frameworks across multiple global client programs, enabling real-time performance visibility and faster operational decision-making.
- Directed project planning, resource coordination, risk management, and stakeholder communication for concurrent technology initiatives, improving delivery predictability and execution.
- Optimized AI data workflows through quality assurance, data preparation, and operational process improvements, supporting reliable machine learning development lifecycles.
- Partnered with cross-functional stakeholders and global clients to resolve delivery challenges, strengthen collaboration, and improve customer satisfaction.

Nov 2017 – May 2021
Nairobi

Product Development Officer/ IT Project Coordinator
Labos Technologies

- Led end-to-end product development from requirements gathering through deployment, delivering customer-focused technology solutions aligned with business objectives.
- Developed product roadmaps, project schedules, budgets, and resource plans across multiple technology initiatives, improving delivery coordination and project visibility.
- Conducted market and customer analysis to prioritize product enhancements, resulting in a flagship feature that contributed approximately 15% revenue growth.
- Coordinated cross-functional teams spanning development, business, and customer stakeholders, accelerating delivery while maintaining alignment with user requirements.
- Mitigated project risks through proactive planning and stakeholder engagement, improving delivery predictability.

Education

Jun 2013 – Jun 2018
Nairobi

Bachelor's of Business and Information Technology
Strathmore University

Feb 2009 – Nov 2012
Kitale

High School Certificate
St Anthony's Boys High School

May 2024 – Present
Nairobi

Masters of Data Science and Analytics
Strathmore University

Select Projects

FlowOps – Real-Time Engineering Operations Platform

Designed and implemented a cloud-native streaming platform on Google Cloud to capture and process GitHub push, pull request, and workflow events in near real time. Built secure webhook ingestion with FastAPI and Cloud Run, event streaming with Pub/Sub, distributed processing with Apache Beam and Dataflow, and analytical storage in BigQuery. Implemented Firestore-based duplicate protection, invalid-event routing, dead-letter handling, monitoring, and Terraform-managed infrastructure.

RoadSafe Lakehouse – Cloud Data Engineering Platform

Designed and implemented a cloud-native data platform using Databricks, PySpark, Delta Lake, Unity Catalog, and Google Cloud Platform. Delivered automated ETL pipelines, workflow orchestration, semantic modeling, data quality validation, audit logging, BigQuery integration, and executive business intelligence dashboards.

Customer IQ (CiQ)

Designed an AI-powered customer intelligence platform that combines customer segmentation, churn prediction, customer lifetime value (CLV), campaign analytics, and predictive insights to help organizations better understand, engage, and grow their customer base.

Data Reliability Copilot

Designed and developed a data reliability platform for profiling datasets, detecting quality issues, scoring data health, generating remediation recommendations, comparing audit versions, and assessing machine learning readiness using FastAPI, Pandas, and Python.